Supplementary Material: Reference Requirements of the Benchmark

Anonymous Author(s)

Anonymous Institution

This table lists every row and atom of the five reference sets (version 3). Rows are the published reference rows and rows added from the same reference document (marked †); each row is followed by its atoms, the scoring units. The reference statements are curator paraphrases; the reference documents' own wording is cited in the benchmark files. A provisional adjudication of reviewed items was applied on 23 October 2026: one duplicate merged, one umbrella atom left out of scoring, two compound atoms split, three evidence labels downgraded and one omitted need restored; each change is logged in the benchmark files.

Columns. Lev.: the atom is about catalog content (c), the profile as a whole (p) or the catalog system (s); only content atoms are scored. *Status* in the source: requirement (req.), proposal (prop.), open issue (open), guidance (guid.) or deferred to future work (def.). *Base*: relation to the pinned DCAT-AP release: restates, tightens, extends, partly covered by, or beyond the base; – for atoms that are not about metadata terms, ? if unresolved. *Raw* and *Cons.*: whether the raw corpus and, for mobilityDCAT-AP and HealthDCAT-AP, the pre-consolidation corpus state the atom: S stated, P in part, U uncertain, N not stated, – no such corpus. These are AI labels with verbatim quotations checked by program; human review is pending. *Sc.*: scored by default (✓); m merged into an equivalent atom of another row (`Req-24.1` into `Req-19.7`); u an umbrella of separately scored atoms (`H-4.1` of `H-16.1`, `H-26.1`, `H-28.1`, `H-36.1`, `H-66.2`).

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
mobilityDCAT-AP NAPCORE requirement analysis (2022), Table 1; base DCAT-AP 2.0.1; test case 40 rows (40 published), 98 atoms, 39 scored (39 requirements), 7 restate the base, 31 stated (37 at least in part); pre-consolidation corpus: 38 (39)							
`Req-1` Use sufficiently expressive, context-aware and modular descriptions with explicit structure, types and rules; distinguish entity identities, preserve references, and support persistent identifiers. *(Main report Table 1, original Req-1, PDF pages 15–19)*							
`Req-1.1`	Make metadata descriptions expressive.	p	req.	–	S	S	
`Req-1.2`	Make metadata descriptions rich.	p	req.	–	S	U	
`Req-1.3`	Describe the data adequately.	p	req.	–	S	S	
`Req-1.4`	Support easy access to the data.	p	req.	–	S	S	
`Req-1.5`	Make metadata descriptions context-sensitive.	p	req.	–	U	S	
`Req-1.6`	Make metadata descriptions modular.	p	req.	–	S	S	
`Req-1.7`	Give metadata descriptions a detailed structure.	p	req.	–	S	S	
`Req-1.8`	Use strong typing and rules in metadata descriptions.	p	req.	–	S	S	
`Req-1.9`	Keep referential integrity.	p	req.	–	S	U	
`Req-1.10`	Enable persistent identifiers.	p	req.	–	S	S	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
Req-2 Make data findable and reusable; provide a basis for NAP metadata databases, describe catalogues, data and their delivery options, and support metadata federation. *(Main report Table 1, original Req-2, PDF pages 15–19)*							
Req-2.1	Make datasets discoverable and searchable.	p	req.	–	S	S	
Req-2.2	Make datasets reusable.	p	req.	–	S	S	
Req-2.3	Serve as a baseline for developing a NAP metadata database.	p	req.	–	S	S	
Req-2.4	Allow a detailed description of the registry (catalogue).	p	req.	–	S	S	
Req-2.5	Allow a detailed description of its datasets.	p	req.	–	S	S	
Req-2.6	Allow a detailed description of their distributions and/or data services.	p	req.	–	S	S	
Req-2.7	Allow aggregation and exchange of metadata.	p	req.	–	S	S	
Req-3 Expose descriptions in forms suitable for people and software. *(Main report Table 1, original Req-3, PDF pages 15–19)*							
Req-3.1	Provide the metadata in a human-readable format.	p	req.	–	S	S	
Req-3.2	Provide the metadata in a machine-readable format.	p	req.	–	S	S	
Req-4 Accommodate descriptions in several languages. *(Main report Table 1, original Req-4, PDF pages 15–19)*							
Req-4.1	Provide metadata information in multiple languages.	p	req.	–	S	S	
Req-5 Use shared vocabularies so participants interpret metadata consistently. *(Main report Table 1, original Req-5, PDF pages 15–19)*							
Req-5.1	Adopt a standardised set of vocabularies to ensure semantic interoperability.	p	req.	–	S	S	
Req-6 Use a common serialization that software agents can process automatically. *(Main report Table 1, original Req-6, PDF pages 15–19)*							
Req-6.1	Adopt a standard format or serialisation, for technological interoperability and automated access by software agents.	p	req.	–	S	S	
Req-7 Publish freely accessible documentation on the Web and, where feasible, provide digitally annotated repository documentation. *(Main report Table 1, original Req-7, PDF pages 15–19)*							
Req-7.1	Make the documentation publicly available on the Web without proprietary licences.	s	req.	–	P	S	
Req-7.2	Where possible, annotate the documentation digitally in a public repository.	s	req.	–	P	P	
Req-8 Establish governance and revision procedures, public issue discussion and release planning, and incorporate resolved changes in later versions. *(Main report Table 1, original Req-8, PDF pages 15–19)*							
Req-8.1	Define governance processes for maintaining the profile.	s	req.	–	S	S	
Req-8.2	Define a revision methodology for the profile.	s	req.	–	S	S	
Req-8.3	Allow public commenting on the profile's development.	s	req.	–	S	S	
Req-8.4	Provide an issues and releases planning tool.	s	req.	–	S	S	
Req-8.5	Have the authors discuss and resolve issues and proposals and incorporate them into follow-up releases.	s	req.	–	S	S	
Req-9 Build on DCAT-AP terms when suitable and follow its extension rules for additions. *(Main report Table 1, original Req-9, PDF pages 15–19)*							
Req-9.1	Reuse DCAT-AP classes and properties where possible.	p	req.	–	S	S	
Req-9.2	Add any further classes and properties according to the DCAT-AP extension rules.	p	req.	–	S	S	
Req-10 Account for subsequent changes to the DCAT-AP baseline. *(Main report Table 1, original Req-10, PDF pages 15–19)*							
Req-10.1	Follow the ongoing evolution of DCAT-AP, considering its developments and revisions.	p	req.	–	S	S	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
Req-11 Review the complete CMC element set against mobility catalogue needs. *(Main report Table 1, original Req-11, PDF pages 15–19)*							
Req-11.1	Consider all CMC metadata elements, to identify metadata that better fit the specific requirements of NAPs.	p	req.	–	S	S	
Req-12 Account for other DCAT-AP profiles and INSPIRE metadata practice. *(Main report Table 1, original Req-12, PDF pages 15–19)*							
Req-12.1	Consider existing DCAT-AP extensions (e.g. GeoDCAT-AP for geospatial data).	p	req.	–	S	S	
Req-12.2	Consider the metadata specifications of the INSPIRE Directive.	p	req.	–	S	S	
Req-13 Preserve interoperability with code lists already deployed by NAP operators. *(Main report Table 1, original Req-13, PDF pages 15–19)*							
Req-13.1	When defining controlled vocabularies, take into account the vocabularies already used in operational NAPs, for interoperability with existing solutions.	p	req.	–	P	S	
Req-14 Describe the varied mobility domain in enough detail to distinguish its growing dataset inventory. *(Main report Table 1, original Req-14, PDF pages 15–19)*							
Req-14.1	Provide exhaustive metadata descriptions for mobility data, given the growing number and variety of mobility datasets.	p	req.	–	P	S	
Req-15 Define a common essential element set while permitting deployment-specific additions. *(Main report Table 1, original Req-15, PDF pages 15–19)*							
Req-15.1	Propose essential metadata elements relevant to any NAP, as a minimum set of metadata for European NAPs.	p	req.	–	S	S	
Req-15.2	Allow further metadata elements in individual NAP deployments.	p	req.	–	S	S	
Req-16 Examine the established CMC/DCAT-AP correspondences to identify unrepresented information. *(Main report Table 1, original Req-16, PDF pages 15–19)*							
Req-16.1	Use the canonical mappings between DCAT-AP and the CMC to identify information that DCAT-AP cannot yet represent.	p	req.	–	P	S	
Req-17 Record creation and revision times, description language, and responsibility for maintaining a metadata record. *(Main report Table 1, original Req-17, PDF pages 15–19)*							
Req-17.1	Record the date and time a metadata record was created or last modified.	c	req.	restates	S	S	✓
Req-17.2	State the language in which the metadata are written.	c	req.	tightens	S	S	✓
Req-17.3	Identify who is responsible for creating and maintaining the metadata.	c	req.	extends	S	S	✓
Req-18 State how the metadata record itself may be reused. *(Main report Table 1, original Req-18, PDF pages 15–19)*							
Req-18.1	Describe the conditions of use of the metadata itself.	c	req.	restates	S	S	✓
Req-19 Characterise the data using a title and synopsis, resource kind, dataset and service classifications including MMTIS service categories, language and payload georeferencing. *(Main report Table 1, original Req-19, PDF pages 15–19)*							
Req-19.1	Give the name of a resource.	c	req.	restates	S	S	✓
Req-19.2	Give a brief description of a resource.	c	req.	restates	S	S	✓
Req-19.3	State the type of resource (dataset or service).	c	req.	restates	S	S	✓
Req-19.4	Give the dataset type category, by logical clustering or by the EC Delegated Regulations.	c	req.	beyond	S	S	✓
Req-19.5	Give the service type category listed in Delegated Regulation (EU) 2017/1926.	c	req.	beyond	P	P	✓
Req-19.6	State the language of a resource.	c	req.	restates	S	S	✓
Req-19.7	Describe the georeferencing method applied within the payload.	c	req.	beyond	S	S	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
Req-20 Indicate which ITS delegated regulations apply to a resource. *(Main report Table 1, original Req-20, PDF pages 15–19)*							
Req-20.1	Indicate which Delegated Regulation (MMTIS, RTTI, SRTI, SSTP) covers a dataset or service.	c	req.	beyond	P	S	✓
Req-21 Coordinate, host, maintain and extend multilingual code lists for describing dataset and service content. *(Main report Table 1, original Req-21, PDF pages 15–19)*							
Req-21.1	Support the coordinated development of a controlled vocabulary for describing dataset and service content.	p	req.	–	S	S	
Req-21.2	Host the controlled vocabulary within the profile.	p	req.	–	P	S	
Req-21.3	Maintain the controlled vocabulary.	p	req.	–	S	S	
Req-21.4	Keep the controlled vocabulary extendable.	p	req.	–	S	S	
Req-21.5	Allow the controlled vocabulary to be translated into local languages, for harmonisation.	p	req.	–	S	S	
Req-22 Represent the interval over which the published data remain valid. *(Main report Table 1, original Req-22, PDF pages 15–19)*							
Req-22.1	Describe the period in which the published data do not expire.	c	req.	extends	S	S	✓
Req-23 Describe spatial extent, applicable regions and network coverage using established functional classifications. *(Main report Table 1, original Req-23, PDF pages 15–19)*							
Req-23.1	Describe the area a dataset covers.	c	req.	tightens	S	S	✓
Req-23.2	Describe the regions in which a dataset is valid.	c	req.	tightens	S	S	✓
Req-23.3	Describe the transport network a dataset considers, following existing, widely applicable functional classifications.	c	req.	beyond	P	S	✓
Req-24 Identify payload location-reference methods and coordinate reference systems. *(Main report Table 1, original Req-24, PDF pages 15–19)*							
Req-24.1	Describe the location referencing methods used in a dataset or service.	c	req.	beyond	S	S	m
Req-24.2	Describe the coordinate reference systems used in a dataset or service.	c	req.	tightens	S	S	✓
Req-25 Identify the transport modes and operators represented by the data. *(Main report Table 1, original Req-25, PDF pages 15–19)*							
Req-25.1	Describe the transport modes a dataset deals with.	c	req.	beyond	S	S	✓
Req-25.2	Describe the transport operators a dataset deals with.	c	req.	beyond	S	S	✓
Req-26 Identify the publishing and owning parties for a dataset. *(Main report Table 1, original Req-26, PDF pages 15–19)*							
Req-26.1	State who publishes a dataset.	c	req.	tightens	S	S	✓
Req-26.2	State who owns a dataset.	c	req.	beyond	S	S	✓
Req-27 Describe contractual, licensing and other conditions on data use. *(Main report Table 1, original Req-27, PDF pages 15–19)*							
Req-27.1	State whether a licence governs the use of the data.	c	req.	tightens	S	S	✓
Req-27.2	State whether a contract governs the use of the data.	c	req.	extends	S	S	✓
Req-27.3	Describe any other condition for using the data.	c	req.	extends	S	S	✓
Req-28 Describe the encoding, syntax, grammar and underlying data model used for access. *(Main report Table 1, original Req-28, PDF pages 15–19)*							
Req-28.1	Describe the encoding of the data for access.	c	req.	beyond	S	S	✓
Req-28.2	Describe the syntax of the data.	c	req.	tightens	S	S	✓
Req-28.3	Describe the grammar of the data.	c	req.	beyond	S	S	✓
Req-28.4	Describe the data model of the data.	c	req.	beyond	S	S	✓
Req-29 Offer schema information and example data so consumers can assess suitability before integration. *(Main report Table 1, original Req-29, PDF pages 15–19)*							
Req-29.1	Describe the data schema, so users can evaluate the data before setting up consumption.	c	req.	beyond	S	S	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
Req-29.2	Provide data samples, so users can evaluate the data before setting up consumption.	c	req.	extends	S	S	✓
Req-30	Distinguish precise data-model versions, publication types and extensions, such as those used in DATEX II, for discovery. *(Main report Table 1, original Req-30, PDF pages 15–19)*						
Req-30.1	Describe precisely the version of the data model a dataset adopts (e.g. the DATEX II version), to allow better searching.	c	req.	beyond	P	S	✓
Req-30.2	Describe the publication type of the data model a dataset adopts (e.g. the DATEX II publication type), to allow better searching.	c	req.	beyond	U	S	✓
Req-30.3	Describe the extensions of the data model a dataset adopts (e.g. DATEX II extensions), to allow better searching.	c	req.	beyond	U	S	✓
Req-31	Record update cadence, quality criteria and the history and status of ITS-regulation compliance assessments. *(Main report Table 1, original Req-31, PDF pages 15–19)*						
Req-31.1	Describe the update rate of a dataset.	c	req.	tightens	S	S	✓
Req-31.2	Describe the quality criteria of a dataset.	c	req.	beyond	S	S	✓
Req-31.3	Describe the history of the procedures that assess compliance with the Delegated Regulations on providing data via a NAP.	c	req.	beyond	P	S	✓
Req-31.4	Describe the status of those compliance-assessment procedures.	c	req.	beyond	S	S	✓
Req-32	Define occurrence limits and obligation levels and explain their consequences for database storage, user interfaces and harvesting, consistently with DCAT-AP. *(Main report Table 1, original Req-32, PDF pages 15–19)*						
Req-32.1	Specify the cardinality of each metadata element.	p	req.	–	S	S	
Req-32.2	Specify the obligation of each metadata element (mandatory, recommended, optional).	p	req.	–	S	S	
Req-32.3	Establish clear rules on these cardinalities for NAP deployments (e.g. their implications for database set-ups, GUI designs and API or harvesting mechanisms).	p	req.	–	P	S	
Req-32.4	Consider the existing DCAT-AP rules on cardinalities.	p	req.	–	S	S	
Req-33	Provide identifier guidance. *(Main report Table 1, original Req-33, PDF pages 15–19)*						
Req-33.1	Define guidelines for identifiers.	p	req.	–	S	S	
Req-34	Define permitted values for metadata elements, using suitable controlled vocabularies. *(Main report Table 1, original Req-34, PDF pages 15–19)*						
Req-34.1	Define the expected value range of each metadata element (for example with controlled vocabularies).	p	req.	–	S	S	
Req-35	Track revisions and versions of catalogue metadata. *(Main report Table 1, original Req-35, PDF pages 15–19)*						
Req-35.1	Track changes of metadata records on a NAP.	c	req.	restates	S	S	✓
Req-35.2	Track different versions of metadata records on a NAP.	c	req.	extends	P	S	✓
Req-36	Enable automated conformance checking of metadata. *(Main report Table 1, original Req-36, PDF pages 15–19)*						
Req-36.1	Support the validation, possibly automatic, of metadata that adopts the profile.	s	req.	–	S	S	
Req-37	Explain element meanings and how to apply them within NAP operations. *(Main report Table 1, original Req-37, PDF pages 15–19)*						
Req-37.1	Provide usage rules in addition to the metadata definitions, explaining the semantics of each metadata element.	p	req.	–	S	S	
Req-37.2	Advise how to relate the metadata elements to NAP processes.	p	req.	–	P	S	
Req-38	Explain allocation of metadata between a dataset or service and its multiple delivery resources. *(Main report Table 1, original Req-38, PDF pages 15–19)*						
Req-38.1	Provide guidelines and solutions for dividing metadata between a dataset or service and its multiple resources.	p	req.	–	S	S	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
`Req-39` Explain exchange of mobility-specific descriptions with other-domain portals, including handling extensions during harvesting. *(Main report Table 1, original Req-39, PDF pages 15–19)*							
`Req-39.1`	Provide guidelines on making NAP-specific metadata interoperable with other data portals (e.g. outside the mobility domain).	s	req.	–	S	S	
`Req-39.2`	Cover in these guidelines how napDCAT-AP extensions are handled when harvesting from other portals.	s	req.	–	P	S	
`Req-40` Give guidance for assuring the quality of metadata received by a NAP. *(Main report Table 1, original Req-40, PDF pages 15–19)*							
`Req-40.1`	Provide guidelines for a quality assurance process that checks the quality of metadata collected on a NAP.	s	req.	–	S	S	

GeoDCAT-AP GeoDCAT-AP 1.0 (2015); base DCAT-AP 1.1; test case

46 rows (7 published), 56 atoms, 37 scored (37 requirements), 16 restate the base, 37 stated (37 at least in part)

`G-1` Provide RDF representations of spatial metadata that conform to DCAT-AP, retain meaning as far as practical and support cross-domain reuse. *(GeoDCAT-AP 1.0 §1.1;4.1)*							
`G-1.1`	Provide a DCAT-AP-conformant representation of geospatial metadata.	p	req.	–	P	–	
`G-1.2`	Provide an RDF representation of geospatial metadata that is as comprehensive as possible and avoids semantic loss.	p	req.	–	S	–	
`G-1.3`	Promote the cross-domain reuse of geospatial metadata.	p	req.	–	P	–	
`G-2` Include every mandatory baseline DCAT-AP element. *(GeoDCAT-AP 1.0 §4.1)*							
`G-2.1`	Include, at least, all mandatory DCAT-AP elements.	p	req.	–	U	–	
`G-3` Use other vocabularies for geospatial information only where the baseline does not cover it. *(GeoDCAT-AP 1.0 §4.1)*							
`G-3.1`	Use vocabularies other than DCAT-AP only for geospatial metadata elements that DCAT-AP does not support.	p	req.	–	P	–	
`G-4` Base alignments on established practice and working-group implementation experience. *(GeoDCAT-AP 1.0 §4.1)*							
`G-4.1`	Base the alignments as far as possible on existing practice, in particular the working group's experience.	p	req.	–	P	–	
`G-4.2`	Avoid a negative impact on existing implementations.	p	req.	–	U	–	
`G-5` Assess any new terms carefully and avoid conflicting with ongoing standardisation. *(GeoDCAT-AP 1.0 §4.1)*							
`G-5.1`	Define new terms only when existing vocabularies offer no suitable candidate, and only after careful assessment.	p	req.	–	P	–	
`G-5.2`	Discard new terms that could conflict with standards under preparation.	p	req.	–	N	–	
`G-6` Cover the ISO core and common INSPIRE metadata elements; exclude only optional material where necessary, with theme-specific extensions outside the common scope. *(GeoDCAT-AP 1.0 §4.2)*							
`G-6.1`	Cover all metadata elements of the core profile of ISO 19115.	p	req.	–	S	–	
`G-6.2`	Cover the INSPIRE metadata elements, leaving out only those not common to all INSPIRE spatial data themes.	p	req.	–	P	–	
`G-6.3`	Exclude only optional elements.	p	req.	–	U	–	
`G-7` Offer an additional RDF publication route while retaining the existing INSPIRE metadata obligations and implementations. *(GeoDCAT-AP 1.0 §1.1)*							
`G-7.1`	Offer an RDF syntax binding as an additional way to implement and publish INSPIRE/ISO metadata on DCAT-AP-conformant portals.	p	req.	–	U	–	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
G-7.2	Leave the INSPIRE metadata regulation and technical guidelines, and the collections maintained under them, unchanged.	p	req.	–	N	–	
G-8†	Identify who is responsible for a metadata record and how to contact them. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.16, II.20)*						
G-8.1	Identify who is responsible for a metadata record and how to contact them.	c	req.	extends	S	–	✓
G-9†	Record the date of each metadata record. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.11, II.21)*						
G-9.1	Record the date of each metadata record.	c	req.	restates	S	–	✓
G-10†	State the language of each metadata record. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.7, II.22)*						
G-10.1	State the language of each metadata record.	c	req.	restates	S	–	✓
G-11†	State the character set of each metadata record. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.19, II.24)*						
G-11.1	State the character set of a metadata record.	c	req.	beyond	S	–	✓
G-12†	Give each metadata record its own identifier. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.17)*						
G-12.1	Give a metadata record a file identifier.	c	req.	extends	S	–	✓
G-13†	State the metadata standard a record follows. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.18)*						
G-13.1	State the metadata standard a record follows.	c	req.	restates	S	–	✓
G-14†	State the version of the metadata standard a record follows. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.18)*						
G-14.1	State the version of the metadata standard a record follows.	c	req.	extends	S	–	✓
G-15†	Give each resource a title. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.1)*						
G-15.1	Give each resource a title.	c	req.	restates	S	–	✓
G-16†	Give the dates of creation, publication and last revision of a resource. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.11)*						
G-16.1	Give the date of creation of a resource.	c	req.	beyond	S	–	✓
G-16.2	Give the date of publication of a resource.	c	req.	restates	S	–	✓
G-16.3	Give the date of last revision of a resource.	c	req.	restates	S	–	✓
G-17†	Describe each resource in a short abstract. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.2)*						
G-17.1	Describe each resource in a short abstract.	c	req.	restates	S	–	✓
G-18†	State the language or languages used in a resource. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.7)*						
G-18.1	State the language or languages used in a resource.	c	req.	restates	S	–	✓
G-19†	Classify each resource by topic category. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.8)*						
G-19.1	Classify each resource by topic category.	c	req.	beyond	S	–	✓
G-20†	Give the geographic extent of a resource as a bounding box or a geographic identifier. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.10)*						
G-20.1	Give the geographic extent of a resource as a bounding box or a geographic identifier.	c	req.	restates	S	–	✓
G-21†	State the character encoding of a dataset. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.24)*						
G-21.1	State the character encoding of a dataset.	c	req.	beyond	S	–	✓
G-22†	Give the temporal extent a resource covers. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.11)*						
G-22.1	Give the temporal extent a resource covers.	c	req.	restates	S	–	✓
G-23†	Describe the lineage of a dataset: how it was produced and processed. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.12)*						
G-23.1	Describe the lineage of a dataset: how it was produced and processed.	c	req.	restates	S	–	✓
G-24†	State how spatial data are represented, e.g. as vector or grid data. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.26)*						
G-24.1	State how spatial data are represented, e.g. as vector or grid data.	c	req.	beyond	S	–	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
G-25†	State the format in which each distribution is encoded. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.25)*						
G-25.1	State the format in which each distribution is encoded.	c	req.	restates	S	–	✓
G-26†	State the spatial resolution of a dataset. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.13)*						
G-26.1	State the spatial resolution of a dataset.	c	req.	beyond	S	–	✓
G-27†	Identify the organisations responsible for a resource and the role each plays. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.16)*						
G-27.1	Identify the organisations responsible for a resource and the role each plays.	c	req.	partly	S	–	✓
G-28†	Give links to the resource or to further information about it. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.4)*						
G-28.1	Give links to the resource or to further information about it.	c	req.	restates	S	–	✓
G-29†	State the coordinate reference system of a dataset. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.23)*						
G-29.1	State the coordinate reference system of a dataset.	c	req.	extends	S	–	✓
G-30†	State the temporal reference system of a dataset. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.23)*						
G-30.1	State the temporal reference system of a dataset.	c	req.	extends	S	–	✓
G-31†	State the specifications a resource conforms to, and whether it conforms. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.14)*						
G-31.1	State the specifications a resource conforms to, and whether it conforms.	c	req.	partly	S	–	✓
G-32†	State whether a resource is a dataset, a dataset series or a service. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.3)*						
G-32.1	State whether a resource is a dataset, a dataset series or a service.	c	req.	partly	S	–	✓
G-33†	State the type of a spatial data service, e.g. discovery, view or download. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.9)*						
G-33.1	State the type of a spatial data service, e.g. discovery, view or download.	c	req.	beyond	S	–	✓
G-34†	Characterise each resource with keywords and name the vocabularies they come from. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.8)*						
G-34.1	Characterise each resource with keywords and name the vocabularies they come from.	c	req.	partly	S	–	✓
G-35†	Link a service to the datasets or dataset series it operates on. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.6)*						
G-35.1	Link a service to the datasets or dataset series it operates on.	c	req.	beyond	S	–	✓
G-36†	Give each resource a unique identifier. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.5)*						
G-36.1	Give each resource a unique identifier.	c	req.	restates	S	–	✓
G-37†	State the conditions for accessing and using a resource. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.15)*						
G-37.1	State the conditions for accessing and using a resource.	c	req.	restates	S	–	✓
G-38†	State any limitations on public access to a resource. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.15)*						
G-38.1	State any limitations on public access to a resource.	c	req.	restates	S	–	✓
G-39†	State how often a resource is maintained and updated. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.27)*						
G-39.1	State how often a resource is maintained and updated.	c	req.	partly	S	–	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
G-40†	Report the topological consistency of a dataset as a data quality result. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.14; §5.3)*						
G-40.1	Report the topological consistency of a dataset as a data quality result.	c	req.	beyond	S	–	✓
G-41†	Report the conceptual and domain consistency of a dataset as data quality results. *(GeoDCAT-AP 1.0 §4.2; Annex I (Table 5); Annex II.14)*						
G-41.1	Report the conceptual consistency of a dataset as a data quality result.	c	req.	beyond	S	–	✓
G-41.2	Report the domain consistency of a dataset as a data quality result.	c	req.	beyond	S	–	✓
G-42†	Make spatial datasets, dataset series and services searchable on general data portals. *(GeoDCAT-AP 1.0 §1; §3)*						
G-42.1	Make spatial datasets, dataset series and services searchable on general data portals.	p	req.	–	P	–	
G-43†	Allow ISO 19115/INSPIRE metadata records to be converted into the RDF representation; conversion back is not required. *(GeoDCAT-AP 1.0 §1.1)*						
G-43.1	Support the conversion of ISO 19115/INSPIRE metadata records into the RDF representation; conversion back is not required.	s	req.	–	U	–	
G-44†	Keep the RDF binding as simple as possible, reusing existing vocabularies and avoiding new terms. *(GeoDCAT-AP 1.0 §1.1)*						
G-44.1	Keep the RDF syntax binding as simple as possible.	p	req.	–	U	–	
G-45†	Reuse the published alignments between EuroVoc, GEMET and the INSPIRE themes. *(GeoDCAT-AP 1.0 §1.1)*						
G-45.1	Reuse the published alignments between EuroVoc, GEMET and the INSPIRE themes.	p	req.	–	U	–	
G-46†	Feed recommendations into the DCAT-AP revision so that DCAT-AP and GeoDCAT-AP stay aligned. *(GeoDCAT-AP 1.0 §1.1)*						
G-46.1	Feed recommendations into the DCAT-AP revision so that DCAT-AP and GeoDCAT-AP stay aligned.	s	req.	–	N	–	

HealthDCAT-AP HealthData@EU Pilot D6.2 (2024); base DCAT-AP 3.0.0; development case

69 rows (16 published), 121 atoms, 89 scored (51 requirements), 13 restate the base, 59 stated (84 at least in part); pre-consolidation corpus: 5 (22)

H-1	Use stable, resolvable identifiers so metadata records and their relationships remain reliably identifiable across catalogues. *(D6.2 §5.2.1.1)*						
H-1.1	Give each dataset metadata record a persistent, dereferenceable identifier (HTTP URI).	c	req.	tightens	P	P	✓
H-1.2	Use persistent, dereferenceable URIs in the properties that link a dataset to other datasets or external resources.	c	req.	tightens	S	U	✓
H-2	Represent catalogue relationships as machine-interpretable links that support a health-data knowledge graph. *(D6.2 §5.2.1.2)*						
H-2.1	Represent the relationships between datasets and other resources (publications, codes) as machine-interpretable links, so the catalogue forms a knowledge graph.	p	req.	–	S	N	
H-3	Have data holders review their dataset descriptions at least once a year for accuracy and currency, and keep track of when each description was last reviewed. *(D6.2 §5.2.1.3)*						
H-3.1	Have data holders check at least once a year that their dataset descriptions are accurate and up to date.	s	req.	–	P	N	
H-3.2	Where catalogues manage the review, record when a description was last reviewed, separately from when it was last modified.	c	prop.	beyond	N	N	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
H-4 Make metadata sufficiently interpretable and informative to support machine-assisted search, including semantic and generative-AI use. *(D6.2 §5.2.2)*							
H-4.1	Provide meaningful free-text descriptions (what, why, how, who) that support natural-language and AI-based search.	c	req.	partly	P	S	u
H-4.2	Annotate datasets with concepts and relationships that users can query semantically (e.g. with SPARQL).	c	req.	beyond	S	N	✓
H-5 Allow dataset discovery to be refined using health categories, access bodies, holder types, personal-data categories, age ranges, population size, quality labels and coding information. *(D6.2 §5.2.3)*							
H-5.1	Classify every dataset by health category, using the mandatory controlled list of EHDS Art. 33 categories, to enable a category facet.	c	req.	beyond	P	P	✓
H-5.2	Let users filter datasets by health data access body.	c	req.	beyond	P	P	✓
H-5.3	Let users filter datasets by publisher type.	c	req.	extends	P	P	✓
H-5.4	Describe the categories of personal data a dataset contains, so users can filter by them.	c	req.	beyond	P	P	✓
H-5.5	State the minimum typical age of the population in a population dataset, as a search filter.	c	req.	beyond	S	P	✓
H-5.6	State the maximum typical age of the population in a population dataset, as a search filter.	c	req.	beyond	S	P	✓
H-5.7	Let users filter datasets by population size.	c	req.	beyond	P	P	✓
H-5.8	Let users filter datasets by quality label.	c	req.	beyond	S	P	✓
H-5.9	Let users filter datasets by the coding values they use.	c	req.	beyond	S	P	✓
H-6 Adapt displayed filters to whether the catalogue query concerns personal or non-personal data. *(D6.2 §5.2.3)*							
H-6.1	Adapt the displayed search filters to whether the user is looking for personal or non-personal data.	s	req.	–	U	N	
H-7 Use semantic annotations and shared coding vocabularies to make health descriptions interoperable and searchable. *(D6.2 §5.2.5)*							
H-7.1	Annotate datasets with shared concepts that have persistent identifiers, so datasets about the same concept can be found across the data space.	c	req.	beyond	S	U	✓
H-7.2	Identify the health themes of a dataset with HTTP URIs from an authoritative vocabulary; mandatory.	c	req.	beyond	P	N	✓
H-7.3	Identify the coding systems a dataset uses with HTTP URIs.	c	req.	beyond	S	N	✓
H-7.4	Identify the code values a dataset uses with HTTP URIs linked to external standards such as ICD-10.	c	req.	beyond	S	N	✓
H-8 Describe at least one safe sample distribution for non-public datasets so users can prepare analyses without accessing sensitive records. *(D6.2 §5.2.6)*							
H-8.1	Describe at least one sample distribution (mock-up, synthetic or anonymised data) for every non-public dataset.	c	req.	tightens	P	S	✓
H-9 Accompany sample data with definitions of its variables in a data dictionary. *(D6.2 §5.2.6)*							
H-9.1	Accompany each sample distribution with a data dictionary that defines all its data elements.	c	req.	beyond	P	N	✓
H-10 Allow applicants to identify the variables they need, supporting data minimisation. *(D6.2 §5.2.6 (pp. 43–44))*							
H-10.1	Let data access applicants list the specific variables they need, in line with data minimisation.	s	req.	–	S	N	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
H-11 Provide usable descriptions of data structures and coding so datasets can be understood and processed consistently. *(D6.2 §5.2.6.1)*							
H-11.1	Describe the data structures of a dataset, where available, in a publicly available and consistent manner.	c	req.	beyond	S	N	✓
H-11.2	Describe the data formats of a dataset, where available, in a publicly available and consistent manner.	c	req.	beyond	S	N	✓
H-11.3	Describe the vocabularies of a dataset, where available, in a publicly available and consistent manner.	c	req.	beyond	S	N	✓
H-11.4	Describe the classification schemes of a dataset, where available, in a publicly available and consistent manner.	c	req.	beyond	S	N	✓
H-11.5	Describe the taxonomies of a dataset, where available, in a publicly available and consistent manner.	c	req.	beyond	P	N	✓
H-11.6	Describe the code lists of a dataset, where available, in a publicly available and consistent manner.	c	req.	beyond	S	N	✓
H-12 Describe logical dataset subdivisions and their relationships where large holdings need to be partitioned. *(D6.2 §5.2.6.2)*							
H-12.1	Describe coherent logical ('virtual') datasets within a large data warehouse, each with its own metadata record.	c	req.	restates	U	U	✓
H-12.2	Embed the breakdown of a data source into collections and datasets in the dataset descriptions.	c	def.	?	P	U	
H-13 State intellectual-property and reuse conditions, including those on sample distributions. *(D6.2 §5.2.6.3)*							
H-13.1	State the conditions for re-use (rights) of every distribution, including sample distributions; mandatory.	c	req.	tightens	P	N	✓
H-13.2	Indicate which parts of a dataset are protected by intellectual property rights.	c	req.	extends	P	N	✓
H-13.3	Justify why those parts need intellectual-property protection.	c	req.	extends	N	N	✓
H-13.4	Where intellectual property rights prevent a sample subset, provide a redacted data dictionary instead.	c	prop.	beyond	N	N	✓
H-14 Link to descriptive analytics that help prospective users assess a dataset. *(D6.2 §5.2.7)*							
H-14.1	Link to analytics resources (dashboards, reports, statistics APIs) that let users assess a dataset without accessing the raw data.	c	req.	beyond	P	S	✓
H-15 Expose quality annotations or labels that help assess fitness for reuse. *(D6.2 §5.2.8)*							
H-15.1	Link each quality or utility annotation (e.g. a quality label or certificate) to the dataset or distribution it assesses.	c	req.	beyond	S	N	✓
H-15.2	State the motivation of each quality annotation.	c	req.	beyond	P	N	✓
H-15.3	Provide the content of each quality annotation as text or as a link to a resource.	c	req.	beyond	P	N	✓
H-15.4	Describe dataset quality with one standardised approach across EU data spaces, aligned with StatDCAT-AP and mobilityDCAT-AP (DQV quality annotations).	p	req.	–	P	N	
H-16 Describe the purpose of data collection and the legal basis for processing personal data. *(D6.2 §5.2.9)*							
H-16.1	Describe the purpose for which the data were collected.	c	req.	beyond	S	S	✓
H-16.2	Identify the legal basis for processing personal data.	c	req.	beyond	S	N	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
H-17†	Describe each dataset's source, scope, main characteristics, the nature of its health data and the conditions for making the data available. *(D6.2 §4 (quoting EHDS Art. 55))*						
H-17.1	Describe the source of each dataset.	c	req.	tightens	S	P	✓
H-17.2	Describe the scope of each dataset.	c	req.	tightens	S	U	✓
H-17.3	Describe the main characteristics of each dataset.	c	req.	restates	S	U	✓
H-17.4	Describe the nature of the electronic health data in each dataset.	c	req.	beyond	S	N	✓
H-17.5	Describe the conditions for making the data available.	c	req.	tightens	S	N	✓
H-18†	Indicate the legislation that applies to a dataset (e.g. the EHDS Regulation), so health catalogues can filter the datasets in its scope. *(D6.2 §4.2; §5.3.1.1; §5.3.1.3 (Applicable legislations))*						
H-18.1	State the legislation that applies to each dataset (e.g. the EHDS Regulation), so catalogues can filter the datasets in its scope.	c	prop.	tightens	P	N	✓
H-19†	Classify each dataset's access rights as public or non-public, so catalogues know whether access needs an application. *(D6.2 §5.3.1.3 (Access rights))*						
H-19.1	Classify each dataset's access rights (public, restricted or non-public), so catalogues know whether access needs an application; NSIP data may only be 'non-public' or 'restricted'.	c	prop.	tightens	P	U	✓
H-20†	Provide at least one distribution for every dataset: an access URL or download link for public data, a landing page of the health data access body for non-public data. *(D6.2 §5.3.1.1; §5.3.1.3 (Access rights; Dataset distribution))*						
H-20.1	Provide at least one distribution for every dataset.	c	prop.	tightens	P	N	✓
H-20.2	For public datasets, give an access URL or download link; for non-public datasets, point the distribution to the landing page of the health data access body or equivalent authority.	c	prop.	tightens	P	N	✓
H-21†	Name a contact point for every dataset, so users can ask about it without working out whom to contact. *(D6.2 §5.3.1.3 (Contact point))*						
H-21.1	Give every dataset a contact point that acts as its helpdesk.	c	prop.	tightens	S	N	✓
H-22†	Identify the health data access body that acts as the access gateway for a non-public or personal dataset. *(D6.2 §5.3.1.3 (Health data access body))*						
H-22.1	Identify the health data access body that acts as access gateway for every dataset with personal or non-public data.	c	prop.	beyond	P	N	✓
H-23†	Indicate whether a dataset contains personal data. *(D6.2 §5.3.1.3 (Data type); §5.3.1.4)*						
H-23.1	Indicate whether a dataset contains personal or non-personal data (data type).	c	prop.	extends	S	U	✓
H-24†	Give each dataset's geographical coverage, preferably from controlled vocabularies, as a filter for populations. *(D6.2 §5.3.1.3 (Geographical coverage))*						
H-24.1	Give each dataset's geographical coverage, preferably from controlled vocabularies, as a filter for populations.	c	prop.	tightens	S	P	✓
H-25†	Characterise each dataset with a few main keywords per language rather than long lists. *(D6.2 §5.3.1.3 (Keyword))*						
H-25.1	Characterise each dataset with a few main keywords per language (10–15) rather than long lists.	c	prop.	tightens	P	N	✓
H-26†	Describe a dataset's lineage: how it was created and modified, and by whom. *(D6.2 §5.3.1.3 (Provenance))*						
H-26.1	Describe a dataset's lineage: how it was created and modified, and by whom.	c	prop.	tightens	S	S	✓
H-27†	Identify the publisher as the data holder accountable for making the dataset accessible. *(D6.2 §5.3.1.3 (Publisher))*						
H-27.1	Identify the publisher as the data holder accountable for making the dataset accessible.	c	prop.	tightens	S	N	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
H-28†	Describe the nature and type of the data holder as context for judging a dataset's credibility. *(D6.2 §5.3.1.3 (Publisher note; Publisher type))*						
H-28.1	Describe the nature of the data holder in free text, as context for judging the credibility of the dataset.	c	prop.	beyond	P	N	✓
H-28.2	Classify the data holder by publisher type, using a controlled vocabulary.	c	prop.	tightens	S	N	✓
H-29†	Tag every dataset in a health catalogue with the EU data theme for health. *(D6.2 §5.3.1.3 (Theme); §5.3.1.4)*						
H-29.1	Tag every dataset in a health catalogue with the EU data theme for health (HEAL).	c	prop.	tightens	P	N	✓
H-30†	Identify the data models and other standards a dataset conforms to (e.g. OMOP), so users can find data ready for their analysis. *(D6.2 §4.1; §5.2.6.2.1; §5.3.1.3 (Conforms to))*						
H-30.1	Identify the data model or other standard a dataset conforms to (e.g. OMOP), so users can find data ready for their analysis.	c	req.	tightens	S	N	✓
H-30.2	Identify those standards with machine-interpretable identifiers (HTTP URIs).	c	req.	tightens	S	N	✓
H-31†	Link to documentation that helps users understand the dataset. *(D6.2 §5.3.1.3 (Documentation))*						
H-31.1	Link to documentation that helps users understand the dataset.	c	prop.	tightens	S	N	✓
H-32†	Link a dataset to related resources and publications that show how it has been used. *(D6.2 §5.2.1.2; §5.3.1.3 (Is referenced by; Related resource); §6.1)*						
H-32.1	Link a dataset to the publications and other resources that reference it.	c	prop.	tightens	S	N	✓
H-32.2	Link a dataset to related resources.	c	prop.	tightens	S	N	✓
H-33†	Identify the datasets a dataset was derived from, and let datasets derived from it refer back to it. *(D6.2 §5.3.1.3 (Source); §6.1)*						
H-33.1	Identify the datasets a dataset was derived from; research datasets derived from a health dataset should cite its metadata identifier.	c	prop.	tightens	S	U	✓
H-34†	Give a landing page that provides access to the dataset and further information. *(D6.2 §5.3.1.3 (Landing page))*						
H-34.1	Give a landing page that provides access to the dataset and further information.	c	prop.	tightens	S	N	✓
H-35†	State the language of a dataset where applicable. *(D6.2 §5.3.1.3 (Language))*						
H-35.1	State the language of a dataset where applicable.	c	prop.	tightens	S	N	✓
H-36†	Describe the population a dataset covers, where it contains one. *(D6.2 §5.3.1.3 (Population coverage))*						
H-36.1	Describe the population a dataset covers, where it contains one.	c	prop.	beyond	S	S	✓
H-37†	State the size of a dataset (number of records, number of unique individuals; byte size of each distribution), so users can judge its value. *(D6.2 §2.3.1.3 and §5.3.1.2 (DGA Art. 8); §5.3.1.3 (Number of records))*						
H-37.1	State the number of records in a dataset.	c	prop.	beyond	S	P	✓
H-37.2	State the number of unique individuals in a dataset.	c	prop.	beyond	S	P	✓
H-37.3	State the byte size of each distribution.	c	prop.	tightens	S	U	✓
H-38†	Give a dataset's temporal coverage, leaving the end open while collection continues. *(D6.2 §5.3.1.3 (Temporal coverage))*						
H-38.1	Give a dataset's temporal coverage, leaving the end open while collection continues.	c	prop.	restates	S	N	✓
H-39†	State a dataset's temporal resolution, to judge whether it can be reused. *(D6.2 §5.3.1.3 (Temporal resolution))*						
H-39.1	State a dataset's temporal resolution, to judge whether it can be reused.	c	prop.	tightens	S	N	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
H-40†	State when a dataset must be deleted (retention period), and keep its record available afterwards. *(D6.2 §5.3.1.3 (Retention period))*						
H-40.1	State the retention period after which a dataset must be deleted.	c	prop.	beyond	P	N	✓
H-40.2	Keep a deleted dataset's metadata record available, because publications may reference it.	s	prop.	–	U	U	
H-41†	Give a version designation where the holder follows a versioning strategy. *(D6.2 §5.3.1.3 (Version))*						
H-41.1	Give a version designation where the holder follows a versioning strategy.	c	prop.	restates	S	N	✓
H-42†	State the spatial resolution where it applies, mainly for geospatial data. *(D6.2 §5.3.1.3 (Spatial resolution))*						
H-42.1	State the spatial resolution where it applies, mainly for geospatial data.	c	prop.	restates	S	N	✓
H-43†	Record when each catalogue entry was listed and last changed, and which original metadata it was created from. *(D6.2 §5.2.1.1)*						
H-43.1	Record when each catalogue entry was listed.	c	req.	restates	S	N	✓
H-43.2	Record when each catalogue entry was last modified.	c	req.	restates	S	N	✓
H-43.3	Identify the original metadata an aggregated catalogue entry was created from.	c	req.	restates	S	U	✓
H-44†	State which application profile a metadata record follows, so aggregators can validate it. *(D6.2 §5.2.1.3)*						
H-44.1	State which application profile a metadata record follows, so aggregators can validate it.	c	req.	restates	S	U	✓
H-45†	Indicate whether a dataset's holder is a trusted data holder, as a search facet. *(D6.2 §5.2.3 (facet list))*						
H-45.1	Indicate whether a dataset's holder is a trusted data holder, as a search facet.	c	prop.	beyond	U	U	✓
H-46†	Provide the information the EHDS data quality and utility label needs: documentation, maturity of quality management, coverage, access and provision times, and data modifications. *(D6.2 §5.2.8 (quoting EHDS Art. 56))*						
H-46.1	For the EHDS data quality and utility label, describe, when applicable, how representative the sampled population is.	c	req.	beyond	S	P	✓
H-46.5	For the EHDS data quality and utility label, state the average time a natural person appears in the dataset.	c	req.	beyond	S	P	✓
H-46.2	For the EHDS data quality and utility label, describe the maturity of the data quality management processes, including review and audit processes and the examination of biases.	c	req.	beyond	S	N	✓
H-46.3	For the EHDS data quality and utility label, state the time between collecting the data and adding them to the dataset.	c	req.	beyond	S	N	✓
H-46.6	For the EHDS data quality and utility label, state the time to provide the data after a data access application is approved.	c	req.	beyond	S	N	✓
H-46.4	For the EHDS data quality and utility label, describe modifications of the data: merging and adding data to an existing dataset, including links with other datasets.	c	req.	beyond	S	P	✓
H-47†	Standardise the minimum metadata elements that health data holders must provide under the EHDS Regulation. *(D6.2 §2.2)*						
H-47.1	Standardise the minimum metadata elements that health data holders must provide under the EHDS Regulation.	p	req.	–	S	N	
H-48†	Remain fully compatible with DCAT-AP. *(D6.2 §2.2)*						
H-48.1	Remain fully compatible with DCAT-AP.	p	req.	–	S	N	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
H-49†	Align the minimum elements for open health data with those of DCAT-AP for High-Value Datasets. *(D6.2 §4.2; §5.3.1.1)*						
H-49.1	Align the minimum elements for open health data with those of DCAT-AP for High-Value Datasets.	p	prop.	–	P	N	
H-50†	Align the minimum elements for non-open, non-personal health data with the metadata of national single information points under the Data Governance Act. *(D6.2 §5.3.1.2)*						
H-50.1	Align the minimum elements for non-open, non-personal health data with the NSIP metadata of the Data Governance Act.	p	prop.	–	P	N	
H-51†	Define stricter, conditional minimum element sets for protected and for sensitive (personal) health data. *(D6.2 §4.2; §5.3.1.4)*						
H-51.1	Define stricter, conditional minimum element sets for protected and for sensitive (personal) health data.	p	req.	–	P	N	
H-52†	Take coded values (health themes, coding systems, code values, standards) as HTTP URIs from one shared, authoritative knowledge base (Wikidata). *(D6.2 §5.2.4; §5.2.5)*						
H-52.1	Take coded values (health themes, coding systems, code values, standards) as HTTP URIs from one shared, authoritative knowledge base (Wikidata).	p	req.	–	P	U	
H-53†	Publish data dictionaries in one harmonised machine-readable format (CSVW), so variable definitions are consistent across catalogues. *(D6.2 §5.2.6.1)*						
H-53.1	Publish data dictionaries in one harmonised machine-readable format (CSVW).	p	req.	–	P	N	
H-54†	Provide descriptions in at least one official EU language, preferably the holder's native language, with language tags. *(D6.2 §6.1.3 (quoting EHDS Art. 55(1a)))*						
H-54.1	Provide dataset descriptions in at least one official EU language.	p	req.	–	P	N	
H-54.2	Encourage data holders to create metadata in their native language.	p	req.	–	P	N	
H-55†	Expose national health catalogues through endpoints that the EU datasets catalogue, single information points and other catalogues can harvest. *(D6.2 §2.3.1.3; §6.2)*						
H-55.1	Expose national health catalogues through endpoints that the EU datasets catalogue, single information points and other catalogues can harvest.	s	req.	–	S	P	
H-56†	Keep the data holder's record as the single source of truth when aggregating: copies refer to the original, and access bodies add a persistent identifier without replacing the original one. *(D6.2 §5.2.1.1)*						
H-56.1	Keep copies of metadata linked to the original record at its source.	s	req.	–	S	P	
H-56.2	Update metadata only at its primary source: the data holder, not the access body, owns and edits it.	s	req.	–	P	U	
H-56.3	Where a record's identifier does not meet the profile's identifier rules, have the access body assign an additional persistent URI while keeping the data holder's original identifier.	s	req.	–	P	P	
H-57†	Detect broken URIs in the links between catalogue resources. *(D6.2 §5.2.1.2)*						
H-57.1	Detect broken URIs in the links between catalogue resources.	s	req.	–	U	N	
H-58†	Measure the maturity of the health data space from its metadata, such as the share of URIs, the level of interoperability and conformity to standards. *(D6.2 §5.2.1.2)*						
H-58.1	Measure the maturity of the health data space from its metadata (share of URIs, interoperability, conformity to standards).	s	req.	–	P	N	
H-59†	Order search facets by their relevance, as shown in the pilot use cases. *(D6.2 §5.2.3)*						
H-59.1	Order search facets by their relevance, as shown in the pilot use cases.	s	req.	–	N	U	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
H-60†	Let applicants attach an analysis script to a data request, to obtain anonymised statistical answers. *(D6.2 §5.2.6 (EHDS Art. 47))*						
H-60.1	Let applicants attach an analysis script to a data request, to obtain anonymised statistical answers.	s	req.	–	P	U	
H-61†	Support data holders in transforming existing metadata into the profile, with a user-friendly multilingual editor that needs no technical expertise. *(D6.2 §6; §6.1)*						
H-61.1	Support data holders in transforming their existing metadata into the profile.	s	req.	–	S	P	
H-61.2	Provide a user-friendly, multilingual metadata editor that needs no technical expertise.	s	req.	–	P	P	
H-62†	Validate metadata against the profile (mandatory elements, data types, controlled vocabularies) before publication and when aggregating. *(D6.2 §6.1.2; §6.2.1)*						
H-62.1	Validate metadata against the profile (mandatory elements, data types, controlled vocabularies) before publication and when aggregating.	s	req.	–	S	U	
H-63†	Assign responsible metadata managers at data holders and access bodies; the data holder, not the access body, owns and edits the metadata. *(D6.2 §6.1)*						
H-63.1	Designate responsible metadata managers at data holders and at access bodies.	s	req.	–	P	N	
H-64†	Let access bodies enrich metadata automatically when indicators show that a dataset falls under the EHDS. *(D6.2 §5.3.1.1)*						
H-64.1	Let access bodies enrich metadata automatically (e.g. add the EHDS as applicable legislation) when indicators show a dataset falls under the EHDS.	s	prop.	–	U	U	
H-65†	Set up revision and maintenance procedures, with clear responsibility, for the profile. *(D6.2 §6.2)*						
H-65.1	Set up revision and maintenance procedures, with clear responsibility, for the profile.	s	prop.	–	U	N	
H-66†	Provide the elements DCAT-AP already makes mandatory: a title and a description for every dataset and an access URL for every distribution. *(D6.2 §5.3.1.1; §5.3.1.2 (NSIP table); §5.3.1.3 (Title; Description))*						
H-66.1	Give every dataset a title.	c	prop.	restates	S	N	✓
H-66.2	Give every dataset a description.	c	prop.	restates	S	N	✓
H-66.3	Give each distribution an access URL that gives access to the data or explains how to request it.	c	prop.	restates	S	N	✓
H-67†	Give each distribution of protected (NSIP) data exactly one file format, and offer other formats as separate distributions. *(D6.2 §5.3.1.2 (NSIP distribution table))*						
H-67.1	State the file format of each distribution; mandatory, with exactly one format per distribution, for protected (NSIP) data.	c	prop.	tightens	P	N	✓
H-67.2	Offer the same data in another format as a separate distribution.	c	prop.	restates	S	N	✓
H-69†	Offer a common harmonised data validator in the infrastructure to check datasets that follow common data models. *(D6.2 §5.2.6.2.1)*						
H-69.1	Offer a common harmonised data validator in the infrastructure to check the quality and consistency of datasets that follow common data models.	s	prop.	–	P	N	
H-68†	State how often a dataset is updated (frequency). *(D6.2 §5.3.1.3 (recommended properties: frequency))*						
H-68.1	State how often a dataset is updated (frequency).	c	prop.	tightens	S	N	✓

StatDCAT-AP StatDCAT-AP 1.0.0 (2016); base DCAT-AP 1.1; development case
22 rows (7 published), 27 atoms, 14 scored (10 requirements), 6 restate the base, 4 stated (8 at least in part)

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
S-1 Expose the dimensions that distinguish observations, including time and demographic or regional dimensions. *(StatDCAT-AP 1.0 §7.2.1)*							
S-1.1	Describe the dimensions that identify observations in a statistical dataset (e.g. time, sex, age, region), the same way whatever the distribution format.	c	req.	beyond	P	–	✓
S-2 Expose attributes needed to interpret observations, such as scaling and observation status. *(StatDCAT-AP 1.0 §7.2.1)*							
S-2.1	Describe the attributes needed to qualify and interpret observed values (e.g. units of measure, scaling factors, observation status), the same way whatever the distribution format.	c	req.	beyond	P	–	✓
S-3 Associate statistical datasets with descriptive quality information. *(StatDCAT-AP 1.0 §7.2.2)*							
S-3.1	Link quality information to the statistical dataset or distribution it describes.	c	req.	beyond	N	–	✓
S-3.2	State the motivation of each quality annotation.	c	req.	beyond	N	–	✓
S-3.3	Provide the quality information as text or as a link to quality information published elsewhere.	c	req.	beyond	N	–	✓
S-4 Identify delivery resources that present dataset visualisations. *(StatDCAT-AP 1.0 §7.2.3)*							
S-4.1	Link to visualisations of the data (tables, graphs or interactive views), modelled as distributions of type visualisation.	c	req.	extends	N	–	✓
S-5 Indicate the actual number of statistical series represented in a dataset. *(StatDCAT-AP 1.0 §7.2.4)*							
S-5.1	State the actual number of data series in a dataset, as distinct from the number of observations and the theoretical number of combinations.	c	req.	beyond	N	–	✓
S-6 Identify the unit in which dataset observations are measured. *(StatDCAT-AP 1.0 §7.2.5)*							
S-6.1	State the unit of measurement of the observations in a dataset.	c	req.	beyond	S	–	✓
S-7 Represent time-series coverage and periodicity without relying on an ambiguous shorthand. *(StatDCAT-AP 1.0 §7.2.6)*							
S-7.1	State the start of the period a time series covers.	c	guid.	restates	S	–	✓
S-7.2	State the end of the period a time series covers.	c	guid.	restates	S	–	✓
S-7.3	State the periodicity of a time series.	c	guid.	restates	U	–	✓
S-8† Improve the discoverability of statistical datasets on general open data portals. *(StatDCAT-AP 1.0.0 §2.2 (charter); §5.1)*							
S-8.1	Improve the discoverability of statistical datasets on general open data portals.	p	req.	–	S	–	
S-9† Let statistical portals feed general open data portals by exporting their metadata in the profile, with guidance for SDMX-based systems. *(StatDCAT-AP 1.0.0 §5.2; §8; Annexes IV–V)*							
S-9.1	Let statistical portals export their metadata in the profile, so general open data portals can take it up.	s	req.	–	S	–	
S-9.2	Give guidance for mapping and extracting metadata from SDMX-based systems.	s	req.	–	U	–	
S-10† Conform fully to DCAT-AP 1.1, meeting every obligation of its conformance statement. *(StatDCAT-AP 1.0.0 §7.1)*							
S-10.1	Conform fully to DCAT-AP and keep statistical descriptions aligned with DCAT and DCAT-AP.	p	req.	–	P	–	
S-11† Concentrate on metadata elements that help users discover datasets. *(StatDCAT-AP 1.0.0 §6.2)*							
S-11.1	Concentrate on the metadata elements that help users discover datasets.	p	req.	–	P	–	

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
S-12†	Take terms from existing, well-known and well-maintained vocabularies, including the ISA Core Vocabularies and Eurostat's metadata vocabularies. *(StatDCAT-AP 1.0.0 §6.2)*						
S-12.1	Take terms from existing, well-known and well-maintained vocabularies.	p	req.	–	S	–	
S-13†	Prefer common controlled vocabularies maintained by the EU Publications Office. *(StatDCAT-AP 1.0.0 §6.2)*						
S-13.1	Prefer common controlled vocabularies, preferably those maintained by the EU Publications Office.	p	req.	–	P	–	
S-14†	Balance simplicity and expressiveness for a wide, non-specialist audience. *(StatDCAT-AP 1.0.0 §6.2)*						
S-14.1	Balance simplicity and expressiveness for a wide, non-specialist audience.	p	req.	–	N	–	
S-15†	Make the profile usable for datasets in any format, such as SDMX, Data Cube or CSV. *(StatDCAT-AP 1.0.0 §7.1; Annex III)*						
S-15.1	Make the profile usable for datasets in any format, such as SDMX, Data Cube or CSV.	p	req.	–	S	–	
S-16†	Classify statistical datasets by EU data theme, mapping Eurostat's own themes onto it. *(StatDCAT-AP 1.0.0 §7.8.3)*						
S-16.1	Classify statistical datasets by EU data theme, mapping Eurostat's own themes onto it.	c	req.	restates	P	–	✓
S-17†	Link an RDF distribution to the data structure definition it follows. *(StatDCAT-AP 1.0.0 Annex III (resolution log))*						
S-17.1	Link an RDF distribution to the data structure definition it follows.	c	guid.	restates	S	–	✓
S-18†	Record publication and modification times to the time of day when data are published more than once a day. *(StatDCAT-AP 1.0.0 §11; Annex III)*						
S-18.1	Record publication and modification times to the time of day when data are published more than once a day.	c	req.	restates	P	–	✓
S-19†	Identify the measures a statistical dataset contains, as SDMX and Data Cube define them. *(StatDCAT-AP 1.0.0 Annex III (resolution log))*						
S-19.1	Identify the measures a statistical dataset contains, as SDMX and Data Cube define them.	c	def.	beyond	S	–	
S-20†	Describe the quality of statistical data in more detail, e.g. following the ESS Single Integrated Metadata Structure (SIMS). *(StatDCAT-AP 1.0.0 §7.2.2; Annex III)*						
S-20.1	Describe the quality of statistical data in more detail, e.g. following the ESS Single Integrated Metadata Structure (SIMS).	c	def.	beyond	N	–	
S-21†	Show how DDI descriptions of datasets can be expressed with the profile. *(StatDCAT-AP 1.0.0 Annex III (resolution log))*						
S-21.1	Show how DDI descriptions of datasets can be expressed with the profile.	p	def.	–	U	–	
S-22†	Document, with a full example, how datasets, distributions, Data Cube datasets and data structure definitions relate. *(StatDCAT-AP 1.0.0 Annex III; Annex VII)*						
S-22.1	Document, with a full example, how datasets, distributions, Data Cube datasets and data structure definitions relate.	p	req.	–	P	–	

DCAT-AP-JRC Perego et al., SDSVoc 2016, paper 27; base DCAT-AP 1.1; development case

11 rows (5 published), 18 atoms, 17 scored (9 requirements), 4 restate the base, 6 stated (12 at least in part)

JRC-1	Identify the people who created a dataset. *(Metadata elements relevant for scientific data, numbered item 1)*						
JRC-1.1	Identify the authors of a dataset.	c	req.	beyond	S	–	✓
JRC-2	Describe the origin and processing history of data. *(Metadata elements relevant for scientific data, numbered item 2)*						
JRC-2.1	Describe the lineage of a dataset.	c	req.	restates	S	–	✓

ID	Reference requirement (atom)	Lev.	Status	Base	Raw	Cons.	Sc.
JRC-3	Give prospective users practical guidance on using a dataset. *(Metadata elements relevant for scientific data, numbered item 3)*						
JRC-3.1	Give potential data consumers information on how to use the data (usage notes).	c	req.	beyond	U	–	✓
JRC-4	Connect datasets to associated scientific publications. *(Metadata elements relevant for scientific data, numbered item 4)*						
JRC-4.1	Link a dataset to scientific publications about it.	c	req.	beyond	P	–	✓
JRC-5	Identify source datasets used in producing a dataset. *(Metadata elements relevant for scientific data, numbered item 5)*						
JRC-5.1	Link a dataset to its input data (the data used to create it).	c	req.	restates	P	–	✓
JRC-6†	Give datasets persistent identifiers, encoded as HTTP URIs where possible, and tell primary identifiers (e.g. DOIs) from alternative ones. *(Perego et al. 2016 (SDSVoc paper 27), "Data citation", "Identifiers")*						
JRC-6.1	Use persistent identifiers for dataset authors and input data, so datasets can be reliably filtered and linked by them.	c	req.	tightens	P	–	✓
JRC-6.2	Encode identifiers as HTTP URIs wherever possible.	c	open	tightens	U	–	✓
JRC-6.3	Distinguish primary identifiers (e.g. DOIs) from alternative ones.	c	req.	restates	S	–	✓
JRC-6.4	Use a DOI as the dataset identifier when generating DataCite citations.	c	req.	tightens	S	–	✓
JRC-6.5	Qualify each identifier with its scheme or type.	c	req.	restates	S	–	✓
JRC-7†	Express the roles that agents play for a dataset beyond publisher and contact point, consistently across catalogues. *(Perego et al. 2016 (SDSVoc paper 27), "Data citation", "Agent roles")*						
JRC-7.1	Use a minimal, consistent set of unambiguous agent roles beyond publisher and contact point.	c	open	beyond	P	–	✓
JRC-8†	Indicate when a distribution gives access through a service or API rather than to the data itself, and describe the service interface with the query parameters needed to select data. *(Perego et al. 2016 (SDSVoc paper 27), "Modelling service/API-based data access")*						
JRC-8.1	Indicate when a distribution points to a service or API rather than directly to the data.	c	open	extends	S	–	✓
JRC-8.2	Describe the API or service interface, with the query parameters needed to select data, so software agents can use it directly.	c	open	beyond	P	–	✓
JRC-9†	Describe provenance in enough detail to support reproducibility and to track how data are used. *(Perego et al. 2016 (SDSVoc paper 27), "Additional requirements" (detailed specification of data provenance))*						
JRC-9.1	Describe provenance in enough detail to support reproducibility and to track how data are used.	c	open	beyond	P	–	✓
JRC-10†	Describe data and metadata quality assessments and include users' feedback on the data. *(Perego et al. 2016 (SDSVoc paper 27), "Additional requirements" (modelling and using data quality assessments))*						
JRC-10.1	Describe data quality assessments of a dataset.	c	open	beyond	U	–	✓
JRC-10.2	Include users' feedback on the data as quality information.	c	open	beyond	U	–	✓
JRC-10.3	Describe quality assessments of the metadata of a dataset.	c	open	beyond	N	–	✓
JRC-11†	Publish metadata on the Web so that the data are visible and discoverable through search engines. *(Perego et al. 2016 (SDSVoc paper 27), "Additional requirements" (optimising metadata publication on the Web))*						
JRC-11.1	Publish metadata on the Web so that the data are visible and discoverable through search engines.	s	open	–	P	–	